

Statistical Modelling and Inference

Week 1

Warm-Up Questions

If a study has only three observations, can we still perform a statistical analysis?

What might go wrong?

A 95% confidence interval means there is a 95% probability that the true parameter lies in the interval.

True or false?

Would a hypothesis test that rejects H_0 every time be a good test?

Why or why not?

Warm-Up Questions

For paired data, an independent two-sample (t)-test is just as appropriate as a paired (t)-test.

True or false?

If a regression model gives ($p > 0.05$), it is reasonable to remove a variable and refit the model until ($p < 0.05$).

True or false?

If 846 out of 20,000 hypothesis tests have ($p < 0.05$), we can conclude that those 846 effects are real.

True or false?

Warm-Up Questions

What do you think?

Suppose I apply to **20 PhD programs**, and each program has an acceptance rate of 20%.

If the decisions were independent, then

$$P(\text{at least one acceptance}) = 1 - (1 - 0.20)^{20} \approx 0.988.$$

So, do I really have a **98.8% chance** of getting into at least one program?

What assumption is being made here? Is it reasonable?

Learning Outcomes for This Week

The purpose of this week's lecture is to ensure that we are on the same page about the *basics*.

- Introduction to statistical models
- Designing studies
- Introduction to simulations
- Procedures of statistical inference

Big Picture: The Statistical Modelling Workflow

Statistical modelling is the process of using data to learn about an unknown population.

- 1. Population**
- 2. Scientific Question**
- 3. Study Design + Data Collection**
- 4. Statistical Model**
- 5. Estimation**
- 6. Uncertainty Quantification**
- 7. Inference + Scientific Conclusions**

Introduction to Statistical Models

Population

In statistics, a **population** is the entire group of individuals or objects we are interested in studying.

We collect a **sample** from the population and use it to make inferences about the population.

Our sample does **not** contain all observations from the population

- Key idea of statistics!
- We have limited information (sample) and use it to learn about the unknown population.
- If the data are a **random sample** from a population, we can make **reasonable inferences** about the population characteristics (e.g. mean)
- The key requirement is that **the sample should be representative of the population**

Population

It is important to clearly define the population of interest.

- it is not enough to say “I collected n observations”
- the key question is: Who or what do these observations represent?

Population

Example 1: Genetic Studies

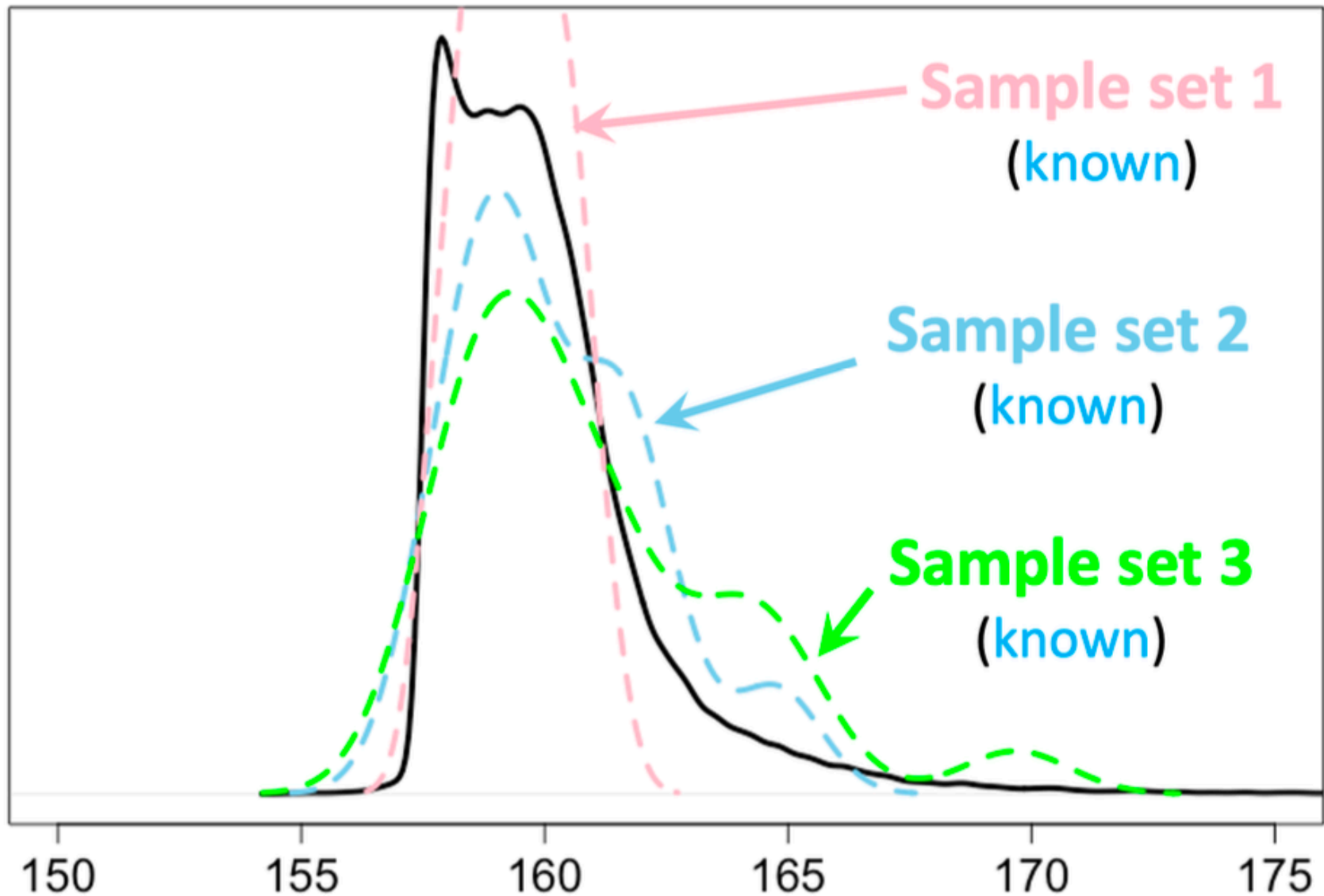
- Many genetic studies have been conducted primarily in people of European ancestry
- Models developed using these data often perform poorly in populations with different genetic backgrounds (e.g., people of Asian or African ancestry).

Example 2: Psychology Experiments

- Psychology studies often recruit undergraduate students because they are readily accessible.
- Do these participants represent all adults?

Population

Heights in cm



- In practice, the true characteristics of a population, such as its mean, variance, or distribution, are typically unknown.

- Our sample provides limited information about the population, but if it is representative, it can provide a good approximation of the population.

Population

- We are often interested in numerical summaries that describe a population.
 - These numerical summaries are called **parameters**
 - e.g. mean height of **all** UofT students
 - Although the parameter is **unknown**, it is a **fixed (constant) value**.
 - The corresponding summaries computed from a sample are called **statistics**.
- Since our conclusions are based on a sample rather than the entire population, every statistical conclusion involves some uncertainty.

Samples

Random sampling is essential for ensuring that study findings generalize to the population of interest.

Example

- In neuroimaging studies, brain MRI scans are often excluded when excessive head motion reduces image quality.
- Adolescents with autism spectrum disorder (ASD) may be more likely to move during scanning and, therefore, more likely to be excluded.
- This can introduce **selection bias**, as the analyzed sample may no longer represent the target population.

Statistical Model

We specify a probability model that represents how we believe the data were generated. This is called the **data-generating process (DGP)** or data-generating model.

- For example, if

$$Y_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{F}$$

then each observation is assumed to be independent and drawn from the same population distribution $\mathcal{F}$.

- One simple model, which makes relatively strong assumptions, is

$$Y_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$$

Examples of Statistical Models

Different research questions require different statistical models!

- (linear) $y_i = \beta_0 + x_i\beta_1 + \epsilon_i$
- (non-linear) $y_i = \beta_0 + \beta_1 \sin(x_i) + \epsilon_i$
- (multiple non-linear) $y_i = f_1(x_{1i}) + f_2(x_{2i}) + \dots + f_p(x_{pi}) + \epsilon_i$
- (multiple non-linear with interactions)
 $y_i = f_1(x_{1i}) + f_2(x_{2i}) + \dots + f_p(x_{pi}) + g_1(f_1(x_{1i})f_2(x_{2i})) + \dots + \epsilon_i$

Statistical Model

- (Binary response)

$$y_i \sim \text{Bernoulli} \left(\frac{\exp(\beta_0 + \beta_1 x_i)}{1 + \exp(\beta_0 + \beta_1 x_i)} \right)$$

Note: this is the default model fit by assuming in `glm(..., family = binomial)` for binary response data

- (Binary data, non linear)

$$y_i \sim \text{Bernoulli} \left(\frac{\exp(\beta_0 + \beta_1 \sin(x_i))}{1 + \exp(\beta_0 + \beta_1 \sin(x_i))} \right)$$

Comments on the Statistical Model

“All models are wrong, but some are useful” (George Box)

- Standard errors, confidence intervals, and hypothesis tests all assume that the statistical model has been correctly specified.
- They do **not** account for errors caused by choosing the wrong model
- Statistics is not assumption-free!
- We make assumptions that we believe are reasonable approximations to reality.
- If these assumptions are substantially violated, the consequences may include:
 - biased estimates
 - incorrect standard errors
 - invalid statistical inference
- Does this make statistical methods useless?
- No, many statistical procedures are surprisingly robust to modest violations of their assumptions.
- We can evaluate through simulations

Comments on the Statistical Model

Example: Suppose we want to compare the mean outcome before and after treatment using paired observations.

For subject $i = 1, \dots, n$, measured at visits $j = 1, 2$, suppose we assume

$$Y_{ij} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_j, \sigma^2)$$

and test $H_0 : \mu_1 = \mu_2$.

What assumption does this model make about repeated measurements from the same subject?

- Is that assumption reasonable?
- What are the consequences if it is violated?

Designing Studies

Design of Studies

1. **Specify the statistical model**

- Propose a DGP that is a reasonable approximation of reality
- Define the research question and hypothesis
- Plan the study design (e.g., sampling strategy and sample size)

2. **Collect the data**

- Obtain a (ideally random) sample from the population of interest

3. **Estimate the quantities of interest**

- Estimate the unknown parameters (e.g., using maximum likelihood estimation)
- Quantify the uncertainty in these estimates (e.g., standard errors or confidence intervals)

4. **Draw statistical conclusions**

- Construct appropriate test statistics or confidence intervals
- Make conclusions about the population based on observed data

Design of Studies

In practice...

- The assumed model does not have to be the true DGP, it only needs to be a useful approximation
- If the sample size is small or estimation is unstable, regularization (penalization) may improve estimation
- If analytic uncertainty quantification is difficult, resampling methods (e.g., the bootstrap) can often be used

Design of Studies

Some cautionary notes

- Ideally, the statistical model and hypotheses are specified **before** examining the data.
- This is informed by background knowledge, scientific theory, and previous research.
- If we repeatedly explore the data until we find an interesting pattern, we may discover relationships that are due to chance rather than true effects.
- A planned analysis may still be inappropriate after data collection.
- **Model assumptions should be checked** after observing the data.
- If assumptions are violated, we may need to modify the analysis
- These decisions should be transparently reported.
- This does not mean we cannot explore data.
- Exploratory analyses can identify new patterns, generate hypotheses, and suggest future research directions.
- Ideally, these findings should be validated using an independent dataset.

Misusing Statistics

Some common practices that can invalidate statistical inference include:

- **P-hacking:** Trying multiple analyses and only reporting one that produces a statistically significant result
- **HARKing (Hypothesizing After the Results are Known):** presenting hypotheses generated from the data as though they were specified before the analysis
- **Repeated Significance Testing:** repeated analyzing the data as participants are recruited and stopping once a significant result is obtained
- **Post-hoc Participant Exclusion:** Modifying inclusion or exclusion criteria after examining the data
- **Using Inappropriate Statistical Methods:** applying statistical tests that do not match the study design, outcome type, or model assumptions
- **Overfitting:** including predictors without scientific justification or appropriate model selection.

Key Point: Statistical inference depends on the entire analysis process. Even a perfectly computed p-value can be misleading if the study design or analysis is flawed.

Misusing Statistics

“If you torture the data long enough, it will confess to anything” (Ronald Coase)

- In principle, the data should be analyzed **only once** to make **confirmatory** results. Any other results would be **exploratory**.
- A good understanding of the scientific domain is essential for designing appropriate studies, selecting sample sizes, specifying hypotheses, and choosing suitable statistical analyses.
- A solid understanding of statistics helps you choose models that provide valid and informative inference.

The goal is not to obtain a small p-value. The goal is to produce evidence that is scientifically credible and statistically valid.

Case Study: Clinical Trials

Developing a new drug requires substantial time and financial investment.

Phase 3 clinical trials provide the evidence needed for regulatory approval. Because these trials are costly and time-consuming, careful statistical planning is essential.

Poor decisions about:

- study design,
- sample size,
- endpoints,
- or statistical analysis

can lead to inconclusive or failed trials, resulting in significant financial losses (i.e., **sunk costs**) and organizational consequences.

Case Study: Clinical Trials

Potential (direct) consequences of sunk cost?

- In 2019, Roche announced plans to cut 500 jobs in Switzerland following the failure of a key clinical trial for its Alzheimer's drug.
- In 2016, The failure of Eli Lilly's Alzheimer's drug, solanezumab, in a Phase 3 trial led to significant restructuring and layoffs.
- Many others!

Case Study: Clinical Trials

A Statistical Analysis Plan (SAP) promotes transparency, reproducibility, and adherence to regulatory requirements.

The SAP is developed before observing trial outcomes to prevent decisions from being influenced by the results.

Key considerations include:

- **Study design:** sample size, randomization, trial structure, and stopping rules.
- **Outcomes:** primary endpoint selection and hypotheses to be tested.
- **Analysis methods:** statistical models, covariate selection, and software.
- **Data handling:** missing data methods and subgroup analyses.
- **Multiple comparisons:** strategies to control false positive findings.

Main Takeaway

- **Statistical inference begins before data collection.**
 - A clear research question, appropriate study design, and reasonable statistical model are essential.
- **Models are approximations of reality.**
 - We do not expect a model to be perfectly true, but it should provide a useful representation of the data-generating process.
- **The analysis process matters.**
 - Statistical conclusions depend on how data are collected, analyzed, and interpreted—not just on the final statistical test.
- **Exploration and confirmation are both valuable, but they serve different purposes.**
 - Exploratory analyses generate hypotheses; confirmatory analyses provide evidence for hypotheses.
- **The goal of statistics is not to find significant results.**

- The goal is to produce reliable, reproducible, and scientifically meaningful evidence.

Introduction to Simulation

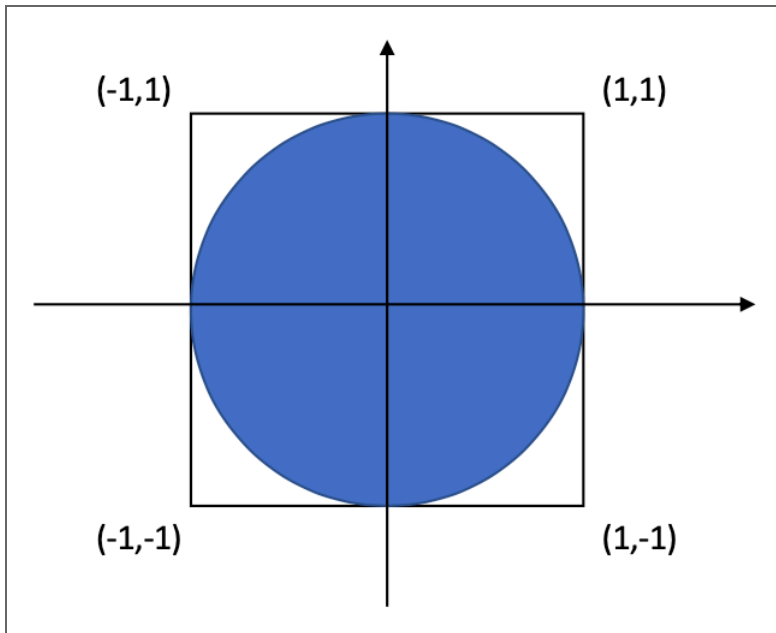
Simulations

Motivating Example

We know from geometry that the area of a unit circle is π .

However, determining the value of π has a long history, requiring increasingly sophisticated mathematical methods.

Modern approaches use calculus to derive and compute this quantity.



Simulations

Motivating Example

How can we estimate the area of a circle using simulation?

1. Generate random points within the square, $[-1, 1] \times [-1, 1]$.
2. The probability that a point falls inside the circle is equal to the ratio of the two areas, $A/4$.
3. Repeat this process many times and estimate the probability by the proportion of points that fall inside the circle.

If M out of 50,000 points fall inside the circle:

$$\frac{A}{4} \approx \frac{M}{50000}$$

Therefore:

$$A \approx 4 \times \frac{M}{50000}$$

Simulations

Motivating Example

R Code [↺ Start Over](#)

[Run Code](#)

```
1  set.seed(123) # Ensure reproducibility
2
3  n <- 50000
4
5  # Generate random points in the square
6  x <- runif(n, -1, 1)
7  y <- runif(n, -1, 1)
8
9  # Calculate distance from origin
10 distance <- sqrt(x^2 + y^2)
11
12 # Identify points inside the unit circle
13 inside <- distance <= 1
14
15 # Estimate pi
16 4 * sum(inside) / n
```

[1] 3.14432

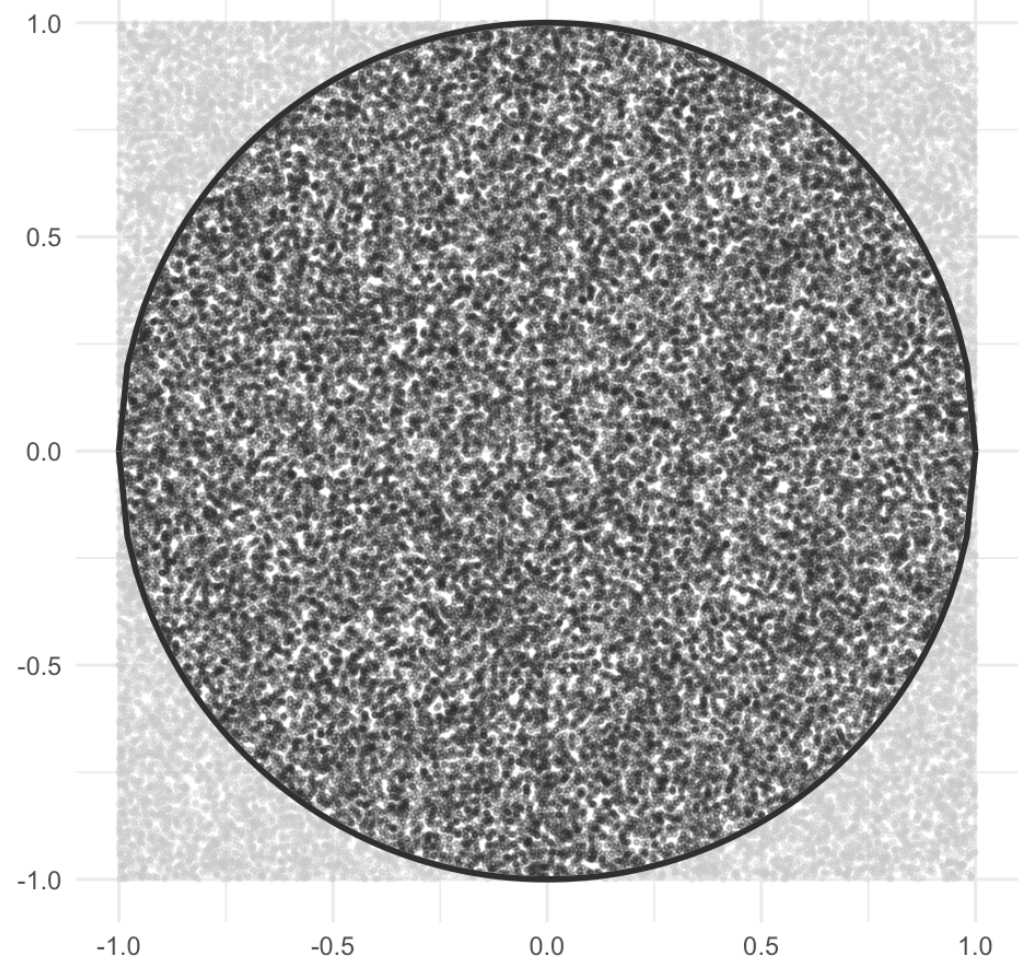
Simulations

Motivating Example

Using 50,000 simulated points:

$$\hat{\pi} = 3.14432$$

- The estimate is not exactly π , but it is close!
- Because the estimate is based on random sampling, it will vary from simulation to simulation.
- Increasing the number of simulated points reduces this variability and gives a more precise estimate.

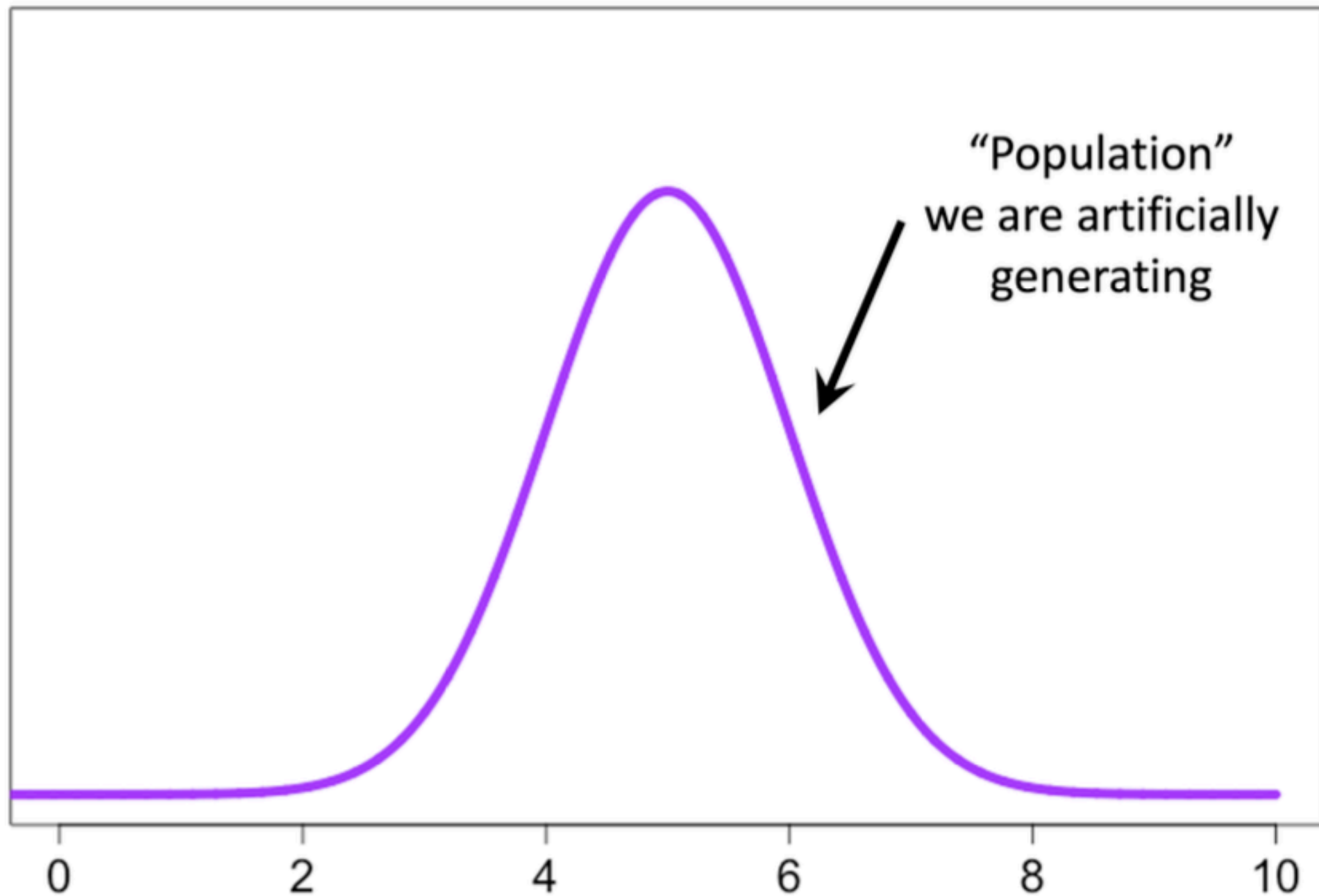


Simulations

Why simulate?

- We can specify a population to generate random samples
 - We then can evaluate statistical properties such as bias of an estimate, errors associated with hypothesis testing, coverage probabilities, etc.
- Later, we will use simulations to see what happens if
- the model is misspecified
 - sample size isn't sufficient
 - statistical misbehaviour is made (e.g. p-hacking)
 - etc.

Simulations



Using simulations, we can generate a population that is bell-shaped, with a fixed mean (e.g., 5) and variance (e.g., 1).

Simulations

A simple example: generating random data in R

R Code

↻ Start Over

Run Code

```
1 ## We want to draw 10 random samples from a population that
2 # follows a normal distribution with mean 5 and standard deviation 1
3 ## We save this data into an object we call 'data'
4
5 data <- rnorm(10, 5, 1)
6 data
```

```
[1] 5.259590 5.917511 4.277682 4.191716 4.858648 7.257013 2.620450 4.545746
[9] 4.939926 5.860901
```

Designing Simulation Studies

A simulation study should be carefully designed to answer a specific statistical question.

A useful framework is **ADEMP** (Morris et al., 2019):

- **A — Aim:** What statistical question are we trying to answer?
- **D — Data-generating mechanism:** How will we generate the simulated data?
- **E — Estimands:** What quantities are we interested in evaluating?
- **M — Methods:** What statistical procedures are being evaluated or compared?
- **P — Performance measures:** How will we assess the methods?

Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11), 2074–2102.

Simulations

Example: Evaluating 'Unbiasedness' Empirically

Assume the following model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \text{where} \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$$

Using the ADEMP framework...

- **A — Aim:** Is the coefficient β_1 of a linear regression model unbiased?
- **D — Data-generating mechanism:** We fix β_0, β_1 , and generate a dataset $(x_i, y_i), i = 1, \dots, n$ where each $X_i \sim \text{Uniform}(0, 5)$. We also generate $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$.
- **E — Estimands:** We estimate β_1
- **M — Methods:** Fitting a linear regression model.
- **P — Performance measures:** Bias (i.e. $E[\hat{\beta}_1] - \beta_1$)

Simulations

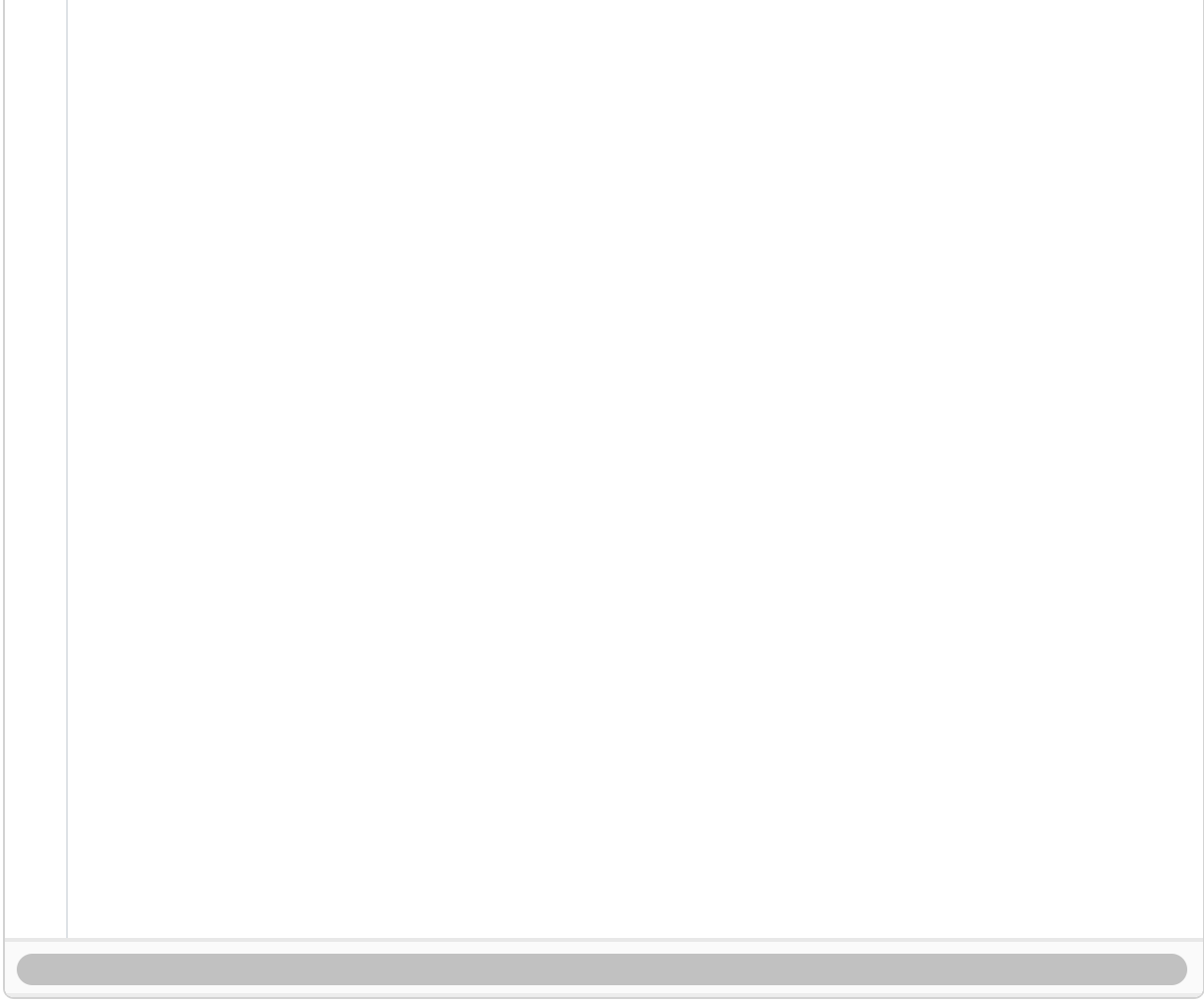
Example: Evaluating 'Unbiasedness' Empirically

R Code

↺ Start Over

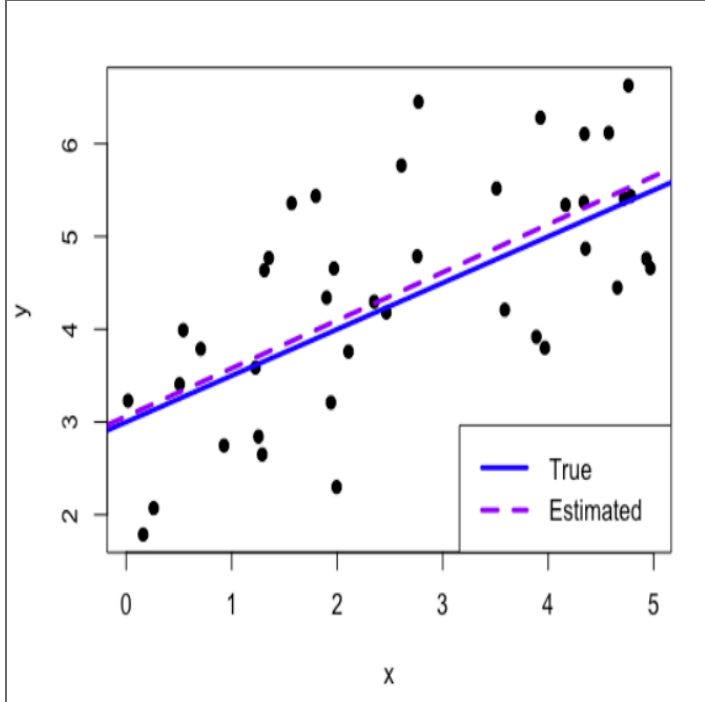
Run Code

```
1  n <- 40
2
3  # Generate the dataset
4  beta0 <- 3
5  beta1 <- 0.5
6
7  x <- runif(n, 0, 5)
8  epsilon <- rnorm(n, 0, 1)
9  y <- beta0+beta1*x + epsilon
10
11 # Fit the model and extract beta1
12 fit <- lm(y~x)
13
14 # Estimated beta_1
15 beta1_hat <- coef(fit)[2]
16 beta1_hat
17
18 # Estimated bias
19
20 beta1_hat - beta1
```



0.6604747^x

0.1604747^x



Simulations

Example: Evaluating 'Unbiasedness' Empirically

Let's repeat this process 1,000 times and calculate the average of all estimated β_1 values to estimate the bias.

R Code [↺ Start Over](#)

[Run Code](#)

```
1  beta1_vec <- NULL
2
3  for (i in 1:1000){
4
5    x <- runif(n, 0, 5)
6    epsilon <- rnorm(n, 0, 1)
7    y <- beta0 + beta1*x + epsilon
8
9    fit <- lm(y~x)
10
11   beta1_vec[i] <- coef(fit)[2]
12
13 }
14
15 # Calculate bias
16
17 mean(beta1_vec) - beta1
```

✓

[1] 0.0008321901

Procedures of Statistical Inference

Hypothesis Testing

In statistics, a **hypothesis** is a statement of one or more parameters

- e.g., $H_0 : E(X_i) = 100$ and $H_A : E(X_i) \neq 100$

The method of **hypothesis testing** uses data from a sample to judge whether a statement about a population may be true or not.

- A simple way you learned is to somehow get a p-value, and compare it to a significance level, α
- If the p-value is less than α , we reject the null hypothesis. Otherwise, we fail to reject the null hypothesis.

Errors in Hypothesis Testing

Since we are making conclusions about the population with limited information, our decision is always prone to some errors.

Type 1 Error: False positive; rejecting the null hypothesis when the null hypothesis is actually true

- $P(\text{Type 1 Error}) = \alpha$

Type 2 Error: False negative; failing to reject the null hypothesis when the null hypothesis is actually false

- $P(\text{Type 2 Error}) = \beta$

Power: the probability that a test will correctly reject a null hypothesis that is false

- Related to Type 2 Error

- $\text{Power} = 1 - P(\text{Type 2 Error}) = 1 - \beta$

Ideally, you want to reduce these two errors, α and β as much as possible.

Errors in Hypothesis Testing

Type I Error

A **Type I error** occurs when we reject H_0 , even though H_0 is true.

$$P(\text{reject } H_0 \mid H_0 \text{ is true}) = \alpha$$

The value α is called the **significance level** of the test.

We choose α in advance, which allows us to control the Type I error rate.

A common choice is

$$\alpha = 0.05.$$

Errors in Hypothesis Testing

Power

The **power** of a hypothesis test is the probability of correctly rejecting H_0 when H_0 is false.

$$P(\text{reject } H_0 \mid H_0 \text{ is false}) = 1 - \beta$$

A test with high power is more likely to detect a real effect when one exists.

Power depends on:

- the **sample size**
- the **effect size**
- the significance level, α

Simulations: Statistical Inference

Case 1: Behaviours of p-values when H_0 is true, $n = 50$

We want to test the following hypothesis:

$$H_0 : \mu = 5 \quad \text{vs} \quad H_A : \mu \neq 5$$

We draw 50 samples from a population that has a mean of 5, i.e., we are setting up a scenario where the **null hypothesis is true**.

Then, using the simulated data, we test the hypothesis.

Simulations: Statistical Inference

Case 1: Behaviours of p-values when H_0 is true, $n = 50$

R Code

↺ Start Over

Run Code

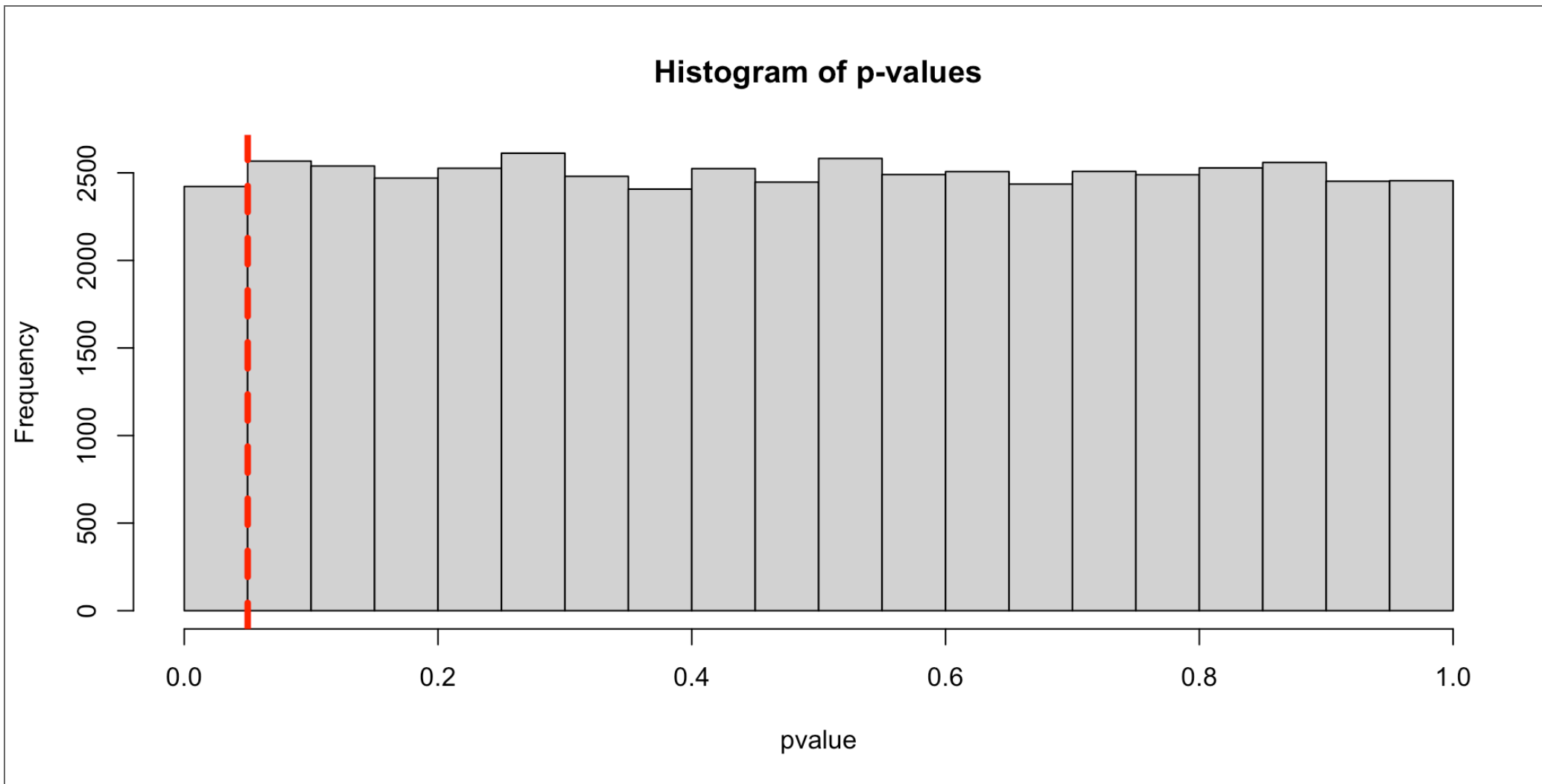
```
1 # Generate 50 observations for a normal distribution with mu = 5
2
3 data <- rnorm(50, 5, 1)
4 # Test the null hypothesis using a t-test
5
6 test <- t.test(data, mu=5)
7 test$p.value
```

[1] 0.6660165

Simulations: Statistical Inference

Case 1: Behaviours of p-values when H_0 is true, $n = 50$

If we repeat this simulation 50,000 times, and then plot a histogram of the resulting p-values, we get the following.



Note that about 5% of the p-values are less than 0.05.

Simulations: Statistical Inference

Case 2: Behaviours of p-values when H_0 is false ($\mu = 5.1$), $n = 50$

R Code

↺ Start Over

Run Code

```
1 # Generate 50 observations for a normal distribution with mu = 5.1
2
3 data <- rnorm(n, 5.1, 1)
4 # Test the null hypothesis using a t-test
5
6 test <- t.test(data, mu=5.1)
7 test$p.value
```

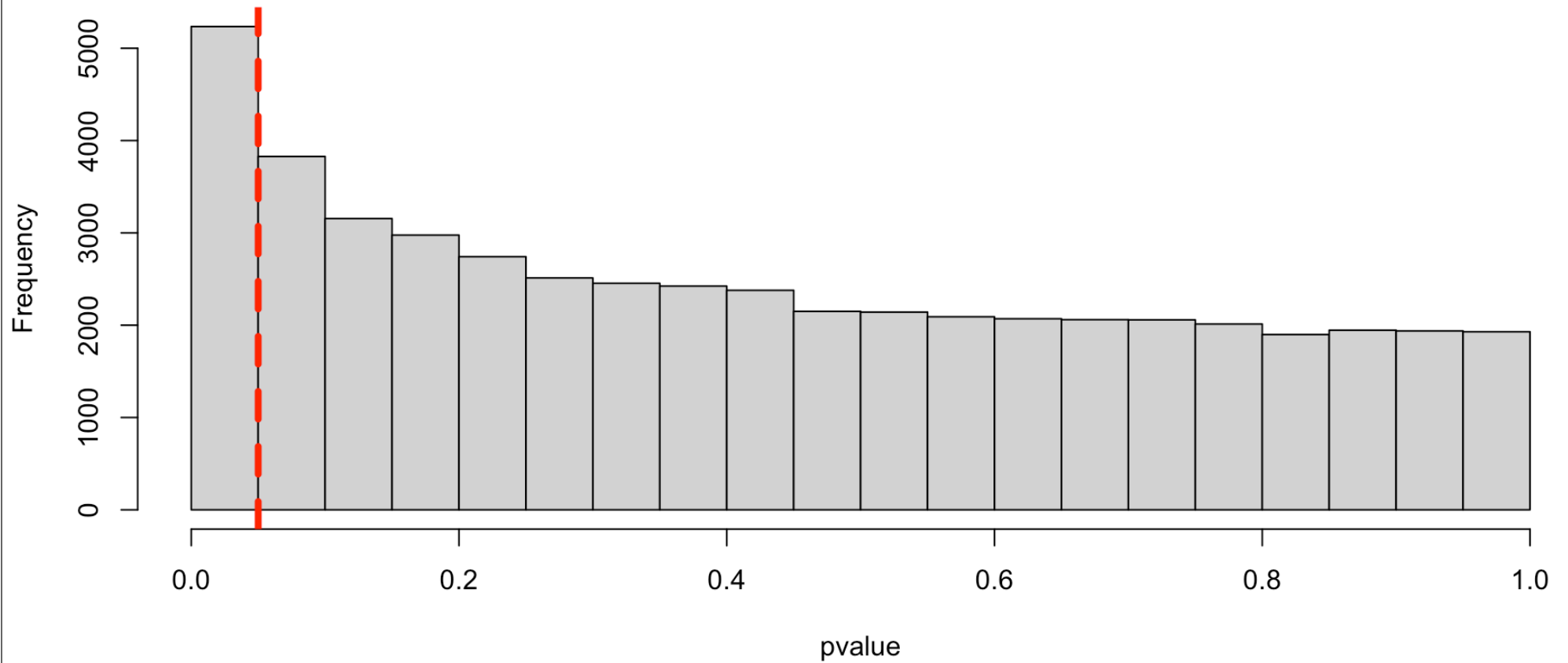
[1] 0.4857103

Simulations: Statistical Inference

Case 2: Behaviours of p-values when H_0 is false ($\mu = 5.1$), $n = 50$

Again, we repeat this simulation 50,000 times - More p-values are shifted to 0 - Is this an expected result?

Histogram of p-values

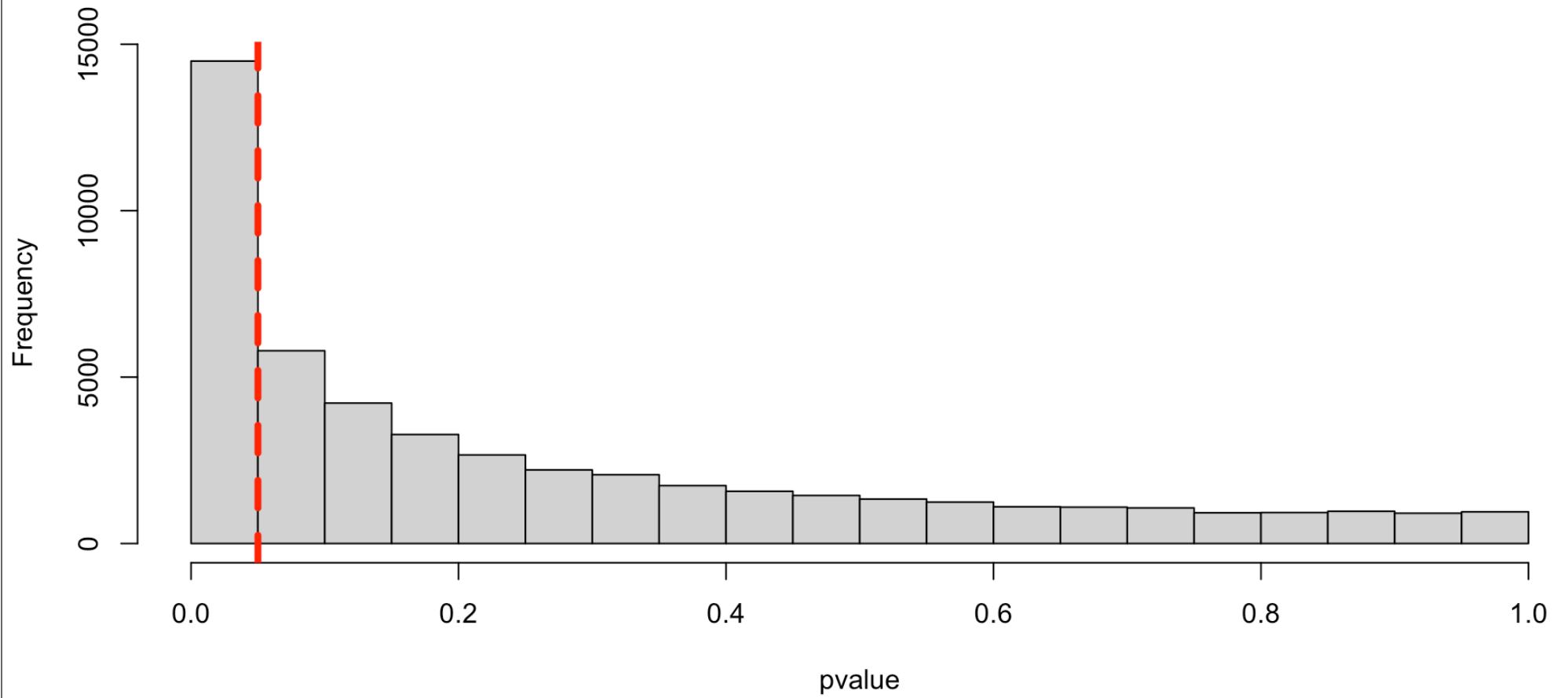


Simulations: Statistical Inference

Case 3: Behaviours of p-values when H_0 is false ($\mu = 5.1$), $n = 200$

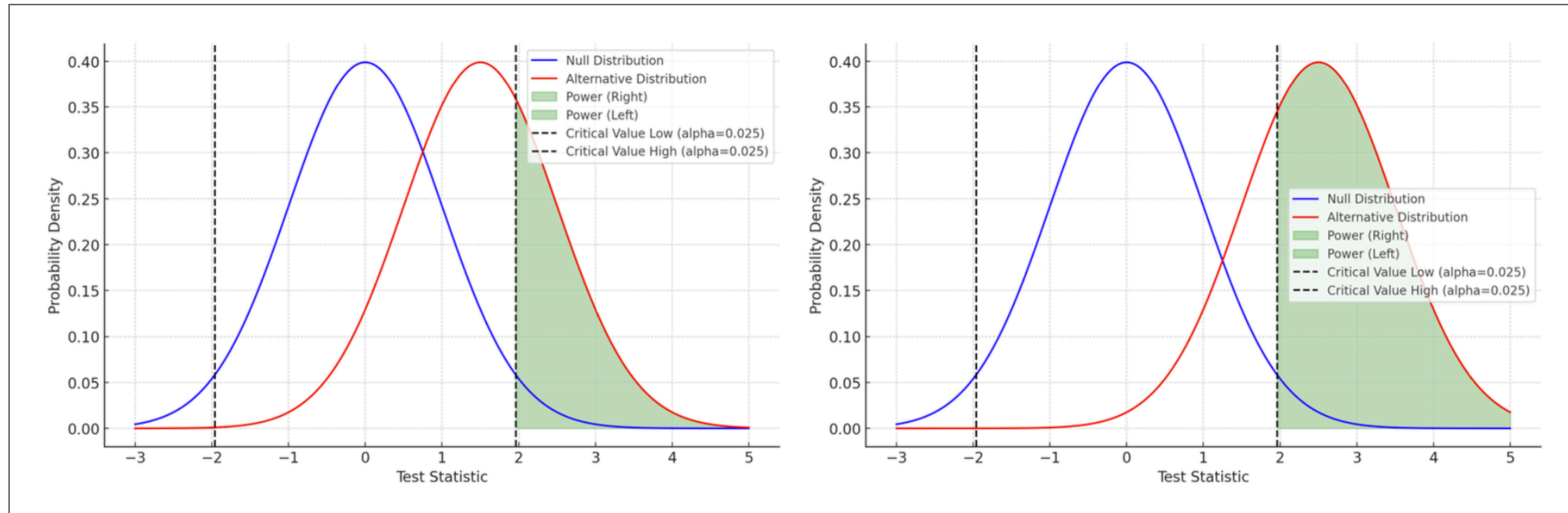
We increase the sample size from $n = 50$ to $n = 200$ and repeat the simulation 50,000 times - More p-values are shifted to 0 - Is this an expected result?

Histogram of p-values



Power

Remember, when talking about power we are assuming the null hypothesis is **false** (i.e., the alternative is true)



Above, we see clearly that α , the effect size and sample size have an effect on power.

Empirical Understanding of Statistics

Simulation lets us ask: *What actually happens?*

We can use simulation to investigate the behaviour of statistical methods when:

- **assumptions are violated**
e.g., non-normal or skewed errors
- **sample sizes are small**
e.g., does the Type I error rate remain at α ?
- **models are misspecified**
e.g., ignoring within-subject correlation
- **we want to evaluate statistical properties**
e.g., bias, power, or confidence interval coverage

Summary

Throughout the course, we will return to the same questions:

What assumptions does the model make?

How do we estimate uncertainty?

How well does the model represent the data-generating process?

We will use **simulations** to investigate the properties of statistical methods and evaluate how well they perform under different scenarios.